Hierarchical Structure Search for Ag/Ag(111) Surfaces

A Four-Stage Pipeline: RSS → Random Assembly → GCGA

Atomistic stack: ASE · pymatgen · spglib · AGOX · GOCIA
Energy model: FAIRChem **UMA-s-1p1** (`task_name="omat"`)
Target system: Ag_{32} clusters on Ag(111) 16×16 supercell (≈1536 atoms)

Working directory: `~/05_AGOX/13`

Motivation & Strategy

Problem. Direct global optimization on a multi-cluster Ag/Ag(111) surface (~1.5k atoms) is intractable: configuration space scales combinatorially with atom count, and full DFT is prohibitively expensive.

Idea. Decompose the search into **physically motivated sub-problems**, each solved with the cheapest method that still captures the relevant physics, then assemble the results.

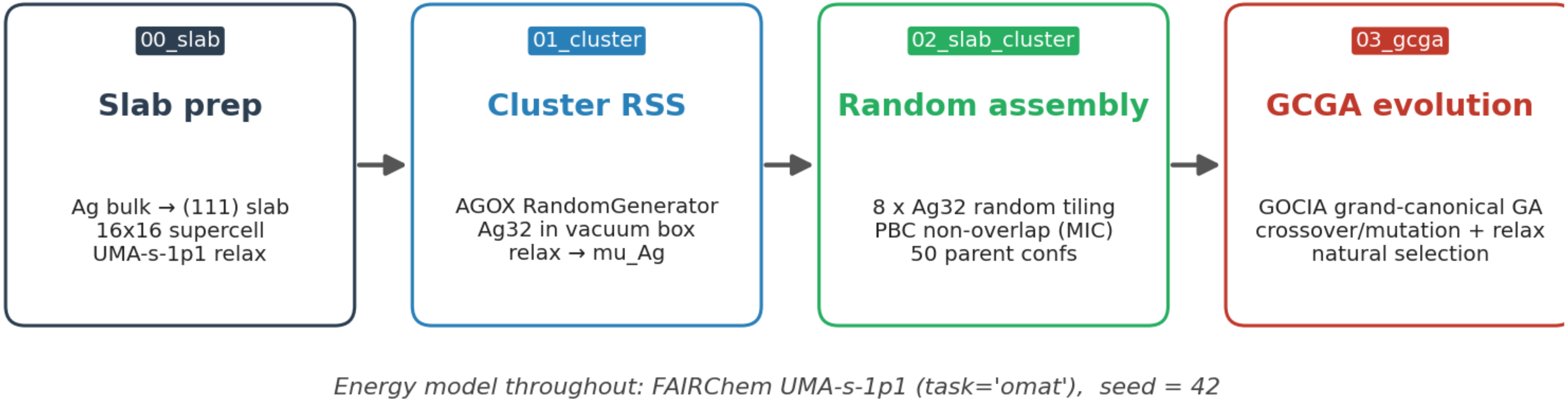
Decomposition

1. **Substrate** — fix the Ag(111) slab once
2. **Building blocks** — find low-energy Ag_{32} motifs in vacuum
3. **Initial population** — randomly tile motifs onto the slab
4. **Global search** — evolve under a grand-canonical GA

Why it works

- Substrate degrees of freedom decoupled from adsorbate
- Cluster RSS explores chemistry of building blocks
- Random tiling seeds GA with diverse, locally sensible parents
- MLIP gives DFT-quality forces at MD-like cost

Pipeline Overview



Single MLIP Every relaxation uses **UMA-s-1p1** / **OMat** for a consistent PES — no cross-method energy bias when seeding the GA.

Seeded RNG `seed = 42` across `random`, `numpy`, `torch` (CPU+CUDA) for reproducibility.

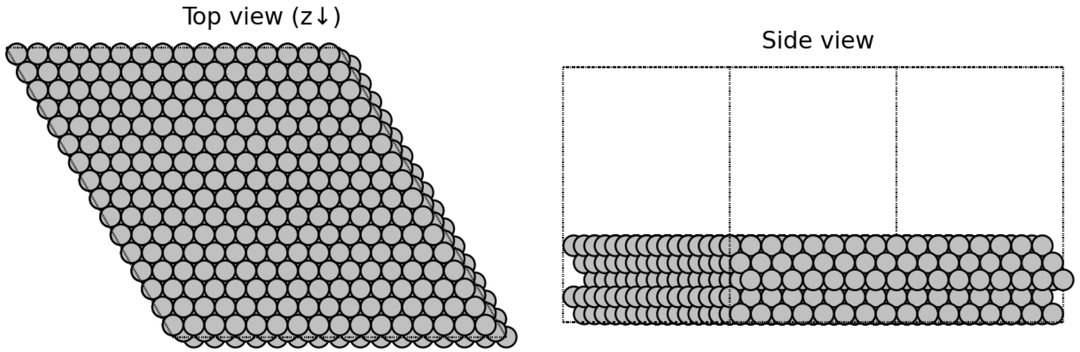
Restart-safe GA loop recovers `kidnum` by scanning `s000NNN/`; `STOP` sentinel halts cleanly.

Stage 1 — Slab Preparation (`00_slab/slab.py`)

Procedure

1. **Bulk vc-relax** of Ag (CIF) with `FrechetCellFilter` + `BFGSLineSearch`, `fmax = 1e-2`
2. **Symmetry-fixed**: `FixSymmetry(symprec=1e-5)` to preserve Fm-3m
3. **Slab cut** with `pymatgen.surface.SlabGenerator`
   - `miller_index = (1,1,1)`
   - `min_slab_size = 8 Å`, `min_vacuum_size = 20 Å`
   - `symmetrize=True`, `lll_reduce=True`
4. **Truncate** topmost atom; **fix bottom two layers** (`FixAtoms`)
5. **Supercell** `slab * (16, 16, 1)` → relax (`fmax = 1e-2`)

Ag(111) 16x16 supercell — 1280 atoms



Output: `4_Ag_slab_supercell_opt.vasp` — 1280-atom substrate reused by every downstream stage. A single fully relaxed substrate avoids re-relaxing it during every GA step.

Stage 2 — Cluster RSS (`01_cluster/`)

`rss_parallel.py` — AGOX random search

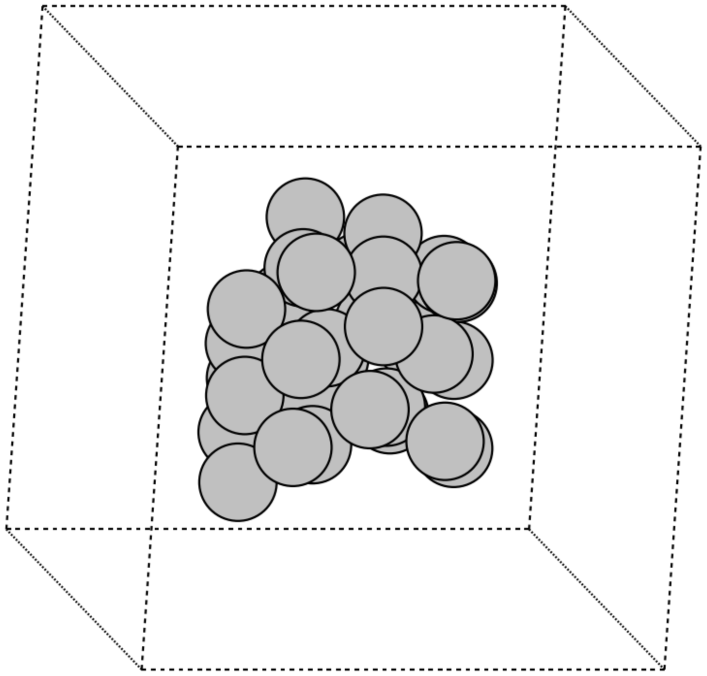
- `RandomGenerator` in a confined cubic box, side $L = (20 n_{\mathrm{Ag}})^{1/3}$ (≈13.5 Å³/atom)
- `LocalOptimizationEvaluator` + `BFGSLineSearch`, `fmax = 1e-2`, `steps = 500`
- **100 AGOX iterations** per stoichiometry; here `n_ag = 32`
- Database `db_Ag32.db` → `all_candidates_Ag32.traj`

Chemical potential

$$\mu_{\mathrm{Ag}} = \frac{1}{N} E_{\mathrm{cluster}}, \quad N = 32$$

The **min** μ_{Ag} (= **–2.084 eV**) seeds the GCGA chemical potential.

Relaxed Ag32 cluster (e.g. relaxed_cluster_000.vasp)



`cluster_relaxation.py` re-relaxes each AGOX candidate under UMA-s-1p1 and reports per-atom energy.

Stage 3 — Slab + Cluster Assembly (`02_slab_cluster/rss_surface.py`)

Geometric placement algorithm

For each of `NUM_STRUCTURES = 50` parents:

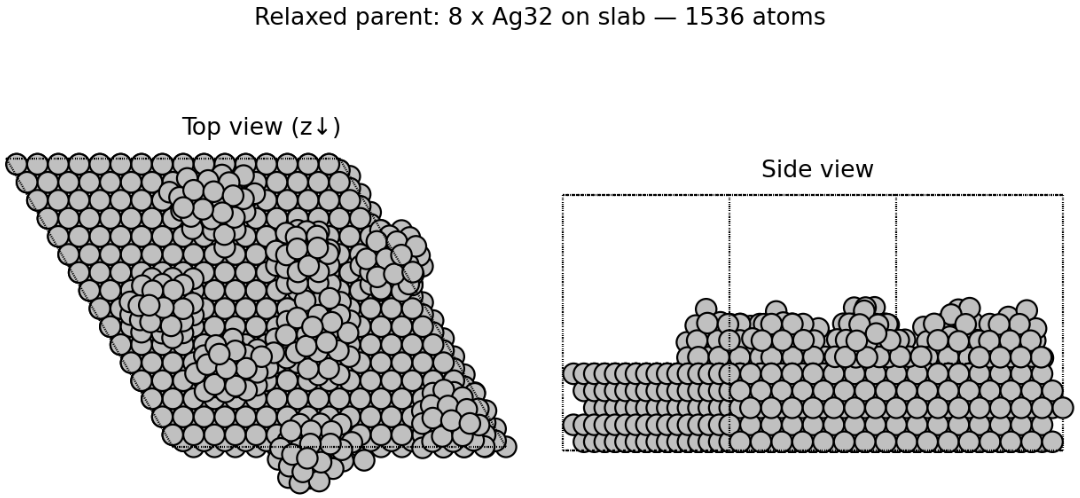
1. Sample **8 distinct** relaxed Ag_{32} clusters
2. For each cluster:
   - center XY at origin, set $z_{\min} = 0$
   - compute enclosing radius $r_i = \max_j \|\mathbf{p}_j^{xy}\|$
3. Draw fractional XY ∈ [0,1]² uniformly
4. **Non-overlap test** with PBC via `ase.geometry.find_mic`:

$$d_{xy}^{\mathrm{MIC}} \geq r_i + r_j + \Delta, \quad \Delta = 3\ \text{Å}$$

5. Place at $z = z_{\mathrm{slab}}^{\max} + 2.5$ Å; retry up to 5000 attempts per cluster, 10 seed-shifted retries per structure

Output

- `generated_surfaces/surface_NN.vasp` — 50 unique configurations
- 1280 (slab) + 256 (8 × Ag_{32}) atoms each



`slab_cluster_relaxation.py` fixes atoms with $z < 11$ Å and relaxes the top with UMA-s-1p1 (`fmax = 0.1`, 1000 steps) → `generated_surfaces_relaxed/`. These seed GCGA.

Stage 4 — Grand-Canonical GA (`03_gcga/`)

Setup — `00_prep_test_trajectory.py`

- Copies `4_Ag_slab_supercell_opt.vasp` → `sub-POSCAR` (first **1280** atoms only)
- Single-point UMA-s-1p1 energy per relaxed `surface_NN.vasp`
- Writes each into `gcga.db` as `parent_*` with `alive=1, done=1`

Evolution — `01_gcga_run.py`

- **GOCIA** `PopulationGrandCanonical`
- Search box: full XY of slab cell, $z \in [8, 25]$ Å
- $\mu_{\mathrm{Ag}} = -2.084104$ eV (from Stage 2)
- `popSize = 50`, `totConf = 3000`, `minConf = 1000`

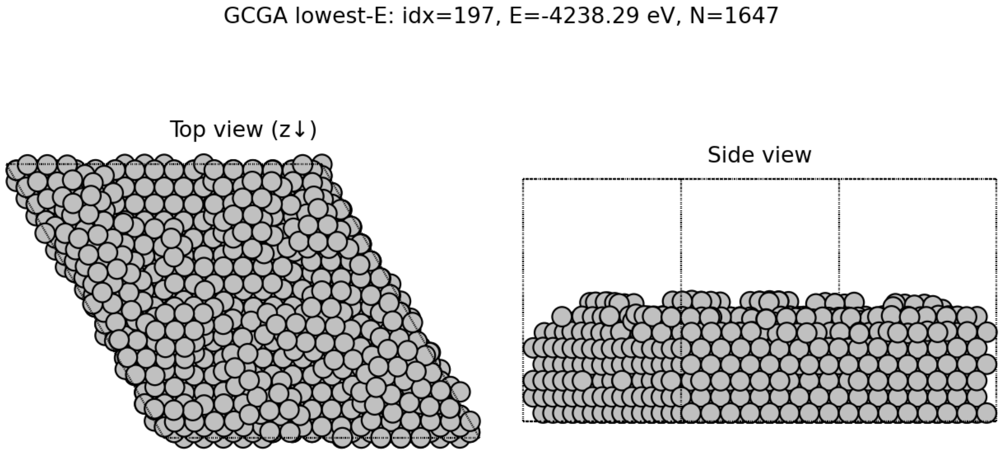
Loop

```
while not converged or kid < minConf:
  kid = pop.gen_offspring_box(
          mutRate=0.5, ...)
  kid.preopt_hooke(cutoff=1.2)
  kid_opt = geomopt_iterate(
      kid, UMA,
      optimizer='BFGSLineSearch',
      fmax=0.1, relax_steps=500,
      substrate='../sub-POSCAR')
  pop.add_aseResult(kid_opt,
                    workdir=kiddir)
  pop.natural_selection()
```

Restart-safe: `kidnum` recovered by scanning `s000NNN/` directories. `STOP` sentinel file halts cleanly.

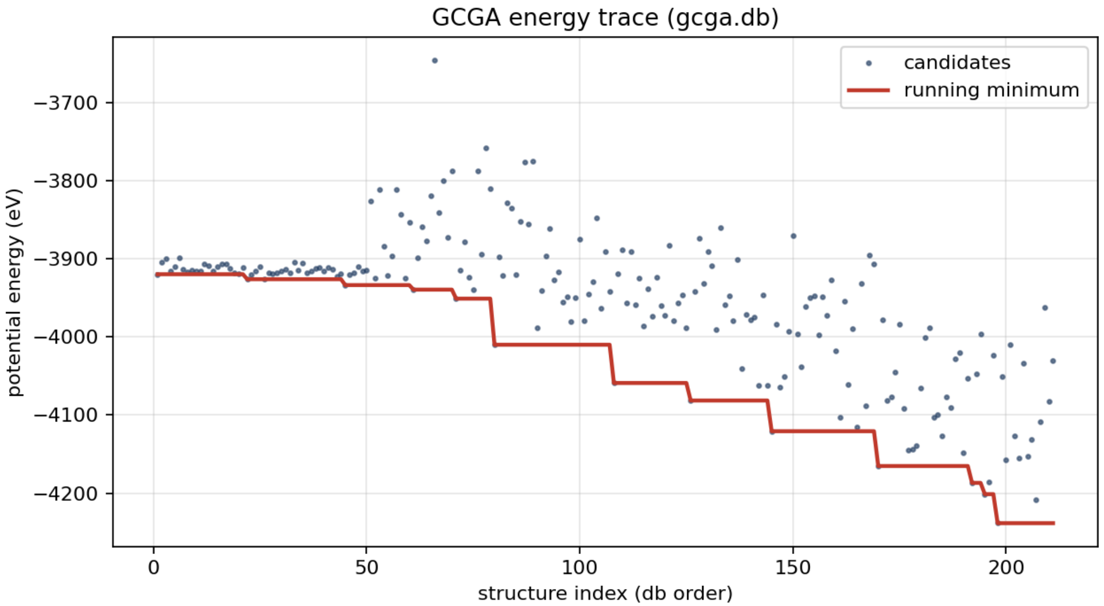
Stage 4 — Current Results (`gcga.db`)

Lowest-energy candidate so far



211 candidates evaluated · running minimum $E = -4238.3$ eV at index 197 · $N_{\text{atoms}} = 1647$

Energy trace



Running-min drops in discrete steps as new low-E motifs survive selection. Scatter spread indicates exploration is still active — GA has not yet stagnated.

Computational Footprint & Hyperparameters

Energy model

Parameter	Value
MLIP	FAIRChem **UMA-s-1p1**
Task	`omat`
Device	CUDA (single GPU)
Bulk fmax	`1e-2` eV/Å
Cluster fmax	`1e-2` eV/Å
Slab+cluster fmax	`1e-1` eV/Å
GA child fmax	`1e-1` eV/Å (≤500 steps)

System sizes

Stage	Atoms	# structures
Bulk	4	1
Cluster	32	100 (RSS)
Slab+cluster	1536	50 parents
GA children	1536 ± Δ	1000–3000

Job scheduler PBS scripts (`batch.sh`, `batch_parallel.sh`) request `select=1:ncpus=20:ngpus=1`. RSS uses a `MAX_JOBS`-throttled background loop; GCGA runs single-threaded over GPU.

Takeaways & Outlook

What the pipeline delivers

- **Reproducible**, MLIP-only path from bulk Ag → globally optimized Ag_x/Ag(111) reconstruction
- Decoupled stages allow swapping components (e.g., different cluster size, different facet)
- μ_{Ag} obtained *in situ* from Stage 2 → consistent grand-canonical reference

Knobs worth exploring

- **Cluster stoichiometry** beyond Ag_{32} (extend the `rss_parallel.py` loop over `n_ag`)
- **Placement density / spacing** Δ in Stage 3 — controls initial population diversity
- **Mutation rate / translation vectors** in `gen_offspring_box` — sets exploration vs. exploitation
- **MLIP refinement**: DFT single-points on GA finalists for publication-quality energies

Validation TODO

- Compare GCGA minima against literature Ag(111) reconstructions
- Convergence diagnostics: $\langle E \rangle$ vs. `kidnum`, population diversity metric